

RESEARCH STATEMENT

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Introduction

During my master's studies with Professor Hsiao, I delved into complex geometry and microlocal analysis, focusing on the Bergman kernel. This tool helps us understand the geometric properties of mathematical spaces by looking at its behavior. My work covers three main areas, with plans to expand them further.

First, we looked at the Bergman kernel's leading behavior, which usually needs positivity and heavy methods. We introduced a flexible "local spectral gap" condition, using scaled coordinates and differential equations to find the main asymptotic behavior. As the parameter k increases, the scaled kernel approaches a quadratic weight. This method, when extended to complex orbifolds, simplifies Tian's 1990 results and suggests uses for *irregular metrics*, guiding my ongoing research.

Second, we establish a simpler way to find the full asymptotic expansion of the Bergman kernel, revealing detailed geometric information. Using the Melin-Sjöstrand stationary phase formula, we compared the kernel to a quadratic model, recovering the expansion and calculating the first two coefficients, which match known results. The leading coefficient involves the determinant of the curvature operator, and the next includes scalar curvatures. This framework may be adaptable to singular or non-compact settings, which I intend to explore further.

Third, we advanced Toeplitz quantization, linking complex geometry to quantization. Beyond functions, we applied it to pseudodifferential operators, including momentum data important for microlocal analysis and physics. Using the stationary phase, we derived a full asymptotic expansion and composition formula, broadening classical results. This opens paths to new quantization frameworks.

Currently, I am extending these results to non-compact or finite-volume manifolds, relaxing previous compact Kähler assumptions, driven by applications in differential geometry and mathematical physics. I am also computing higher Bergman kernel coefficients, which encode geometric data relevant to Donaldson's conjectures. This involves adapting stationary phase methods to control boundary behavior, precisely handling operators and curvature terms, and formulating semiclassical quantization for pseudodifferential Toeplitz operators, where Poisson brackets emerge early. These efforts generalize past findings and lay the groundwork for studying kernels and operators in CR, contact, and sub-Riemannian geometries.

My future research focuses on three areas to deepen these contributions. First, I will explore Szegő kernels in Cauchy-Riemann (CR) geometry, which, like Bergman kernels, reproduce functions and encode curvature and spectral data on pseudoconvex manifolds. Building on my Bergman work, I aim to derive explicit asymptotics and quantization for Szegő kernels, starting with the Heisenberg group using semiclassical techniques. This involves constructing parametrices for the Kohn Laplacian and extending results to general CR manifolds, refining stationary phase methods to capture full expansions. Long-term, I will integrate microlocal pseudodifferential and Fourier integral operators to explore CR quantization, addressing coefficient recurrences and their geometric significance.

Second, I will develop Szegő- and Toeplitz-like operators in contact and conformal geometries, extending complex and CR ideas to broader settings. In contact geometry, I will generalize to non-CR manifolds, adapting Boutet de Monvel's Kohn Laplacian parametrix

to sub-Riemannian Laplacians and exploring symbolic calculus, heat kernels, and spectral properties for new invariants. In conformal geometry, I will construct invariant kernels, study their semiclassical expansions, and analyze coefficients under conformal changes to uncover new invariants, potentially linking to anomalies and critical metrics.

Third, I aim to unify microlocal and symbolic calculus across complex, CR, almost complex/CR, contact, and conformal settings, focusing on operators like Kohn or sub-Riemannian Laplacians. Inspired by Hörmander and Boutet de Monvel, I will develop a tailored calculus for asymptotics, commutators, and parametrices on almost complex/CR or contact settings, extending Donaldson-Sun theory microlocally. This will reveal spectral and symbolic phenomena, connecting kernel expansions to Poisson/symplectic structures and applying Fourier integral techniques to deformation and geometric quantization.

These directions bridge analysis, geometry, and quantization, aiming to uncover new invariants and deepen insights in mathematics and physics.

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1 Past Research Contributions

During my master's studies [Tsa25], under the supervision of Prof. Hsiao, I focused on complex geometry and microlocal analysis, developing methods for the asymptotics of Bergman kernels and for Toeplitz quantization. The Bergman kernel is a central object in complex analysis and complex geometry, encoding the geometry of a line bundle and the underlying manifold through its asymptotic behavior. Its leading term in the high-tensor power limit governs, for instance, the induced Kähler metric, the density of quantum states, and has implications for problems in random complex geometry.

Below, I summarize three representative topics illustrating my contributions. The first is about the leading behavior of the Bergman kernel, which plays an important role in either complex analysis or complex geometry. The second one is the full expansion, which develops the understanding of geometric structures and their meaning. The last one is the Toeplitz quantization, which makes a bridge of microlocal analysis to bring many tools to complex geometric objects.

1.1 Leading Behavior of Bergman Kernel

Traditionally, obtaining this leading behavior requires positivity assumptions on the line bundle and intricate microlocal machinery. We were motivated to ask whether one could recover the same leading term under weaker assumptions, using a more elementary method.

In my Master's work, we introduced the local spectral gap condition, a hypothesis which already contains the positivity of the line bundle but is formulated in a more flexible way. Working in a scaled local coordinate system and treating the problem from a partial differential equation perspective, we derived the leading asymptotic of the Bergman kernel directly.

Theorem 1.1. *Let (M, L, h^L) be a triple consisting of a complex manifold, a line bundle, and a Hermitian metric that satisfies the local spectral gap on an open neighborhood U of $p \in M$. Then we have the leading behavior is given by the quadratic term of the local weight near p . That is*

$$\frac{1}{k^n} P_k \left(\frac{x}{\sqrt{k}}, \frac{y}{\sqrt{k}} \right) = B_{k\phi_0}(x, y) + o(1)$$

as k tends to infinity, where ϕ_0 is the quadratic term of the local weight of L with respect to h^L and the coordinate.

Building on this approach, we extended the local spectral gap framework and the leading term asymptotic to complex orbifolds equipped with orbifold line bundles, providing a unified treatment in this more singular setting.

Theorem 1.2. *Let (X, L, h^L) be a triple of a complex manifold, not necessarily compact, line bundle, and Hermitian metric that satisfies the C_k local spectral gap on an open neighborhood U of $p \in X$. Then we have the leading behavior is given by the quadratic term of the local weight near p . That is*

$$P_k(x, y) = \sum_{g \in G_p} B_{k\phi_0}(\sqrt{k}\tilde{x}, \sqrt{k}(g \cdot \tilde{y})) + \varepsilon_k$$

as k tends to infinity, where

$$\left\| \varepsilon_k \left(\frac{\tilde{x}}{\sqrt{k}}, \frac{\tilde{y}}{\sqrt{k}} \right) \right\|_{C^\ell} = o(k^n),$$

G_p is the isotropic group, $\sim$ means the natural projection, and ϕ_0 is the quadratic term of the local weight of L with respect to h^L and the coordinate.

Although the leading term of the Bergman kernel has been known since Tian's pioneering work [Tia90], our contribution provides an alternative and arguably simpler route that may serve as a stepping stone to further developments. For example, the method suggests a way to treat singular metrics, degenerating sequences of line bundles, or non-standard boundary conditions. These directions form part of the motivation for my current and future work on the interface between complex geometry and microlocal analysis.

1.2 Stationary Phase Method of Bergman Kernel

The full asymptotic expansion of the Bergman kernel is a cornerstone result in complex geometry: each coefficient reflects intricate geometric data of the manifold and line bundle, and the expansion underpins topics such as the quantization of Kähler manifolds and the study of random holomorphic sections.

When the line bundle is positive, historically, Catlin [Cat99] and Zelditch [Zel98] obtained the full expansion via the Szegő kernel, while Dai-Liu-Ma [DLM06] and Marinescu [MM06] later produced proofs using heat kernel and spin^c Dirac techniques. These approaches are powerful but rely on heavy microlocal or spectral analysis. When the line bundle is semi-positive, Marinescu and Savale [MS24] obtained the expansion on Riemann surfaces under the assumption that the curvature vanishes to finite order.

Motivated by our work on the leading term, we developed a more elementary approach based on the Melin-Sjöstrand stationary phase formula [MS75]. By comparing the Bergman kernel with a quadratic-weighted model in the Euclidean setting and working with adjoint operators under two different inner products, we recovered the full asymptotic expansion and explicitly computed the first two coefficients. Our computations coincide with known values [Lu00; Hsi12; MM12], confirming the validity of the method, and the procedure is systematic enough to suggest how to obtain higher-order terms.

Theorem 1.3. *Let M be a compact manifold and (L, h^L) be a positive Hermitian line bundle. There exist $b_j \in \mathcal{C}^\infty(M)$ such that*

$$P_k(z, z) \sim \sum_{j=0}^{\infty} k^{n-j} b_j(z).$$

Here,

$$b_0(z) = \det \left(\frac{\dot{R}^L}{2\pi} \right) (z)$$

and

$$b_1(z) = \det \left(\frac{\dot{R}^L}{2\pi} \right) (z) \left(\frac{1}{4\pi} \hat{r}(z) - \frac{1}{8\pi} r(z) \right)$$

where $\dot{R}^L$ is the curvature operator and $r, \hat{r}$ are scalar curvatures with respect to the manifold and the line bundle, respectively.

Although our current computations stop at the first few coefficients, this stationary-phase framework offers a new tool to investigate the deeper geometric meaning of the coefficients and their relation to global invariants. It also points towards possible extensions to singular metrics, non-compact settings, or to kernels on *CR* and *contact* manifolds, which will be an important focus of my future research.

1.3 Toeplitz Quantization of Pseudodifferential Operators

The classical Toeplitz-Berezin quantization is a key link between complex geometry and quantization: it takes a smooth function on a Kähler manifold, sandwiches it between two Bergman projections, and produces an operator whose asymptotics encodes the underlying symplectic structure. This operator model complements Kontsevich's deformation quantization [Kon03] on the algebraic side.

However, almost all the existing results deal only with multiplication by functions. Little is known about a fully developed Toeplitz framework for pseudodifferential operators, which naturally carry both position and momentum information and are central in microlocal analysis and theoretical physics.

In my Master's work, we extended the Toeplitz framework to classical pseudodifferential operators acting on sections of positive line bundles. Using the stationary-phase approach developed earlier, we derived a full asymptotic expansion for such Toeplitz operators and proved a composition formula, which recovers the usual Toeplitz-Berezin quantization as a special case.

Before proceeding further, we pause to clarify the definition of the Toeplitz operator we will use. Denote by P_k the Bergman projection from the space of square-integrable sections to the subspace of holomorphic square-integrable sections. For a classical pseudodifferential operator $\mathcal{P}$, the associated Toeplitz operator is then defined by $T_{\mathcal{P},k} = P_k \circ \mathcal{P} \circ P_k$. Such operators act on the holomorphic subspace and form the analytic backbone of Toeplitz quantization.

To the best of my knowledge, this is the first systematic extension of the Toeplitz quantization framework to pseudodifferential operators, providing a microlocal description of quantization directly on complex manifolds.

Theorem 1.4. *Let $\mathcal{P}$ be a classical pseudodifferential operator of order m with symbol $p \sim \sum_{j=0}^{\infty} p_j$. There is a sequence of smooth functions $t_{p,j}$ such that*

$$T_{\mathcal{P},k}(z, z) \sim \sum_{j=0}^{\infty} k^{n+m-j} t_{p,j}(z).$$

Here,

$$t_{p,0}(z) = b_0(z) p_0(z, -Jd\phi(z))$$

and

$$t_{p,1}(z) = b_1(z) p_0(z, -Jd\phi(z)) + b_0(z) \left[\sigma_{\text{sub}}(\mathcal{P})(z, -Jd\phi(z)) + \frac{1}{2} \Delta_L(p_0(z, -Jd\phi(z))) + \hat{\sigma}_{p_0}(z, -Jd\phi(z)) \right]$$

where σ_{sub} denotes the subprincipal symbol. The detailed definition of these geometric invariants could be found in section 5.3 in my thesis [Tsa25].

Theorem 1.5. *Let $\mathcal{P}$ and $\mathcal{Q}$ be classical pseudodifferential operators of order m_1 and m_2 , and with symbols p and q , respectively. There exists a sequence $C_j(\mathcal{P}, \mathcal{Q}; k)$ of classical pseudodifferential operators with order $m_1 + m_2$ such that*

$$T_{\mathcal{P},k} \circ T_{\mathcal{Q},k} \sim \sum_{j=0}^{\infty} k^{-j} T_{C_j(\mathcal{P}, \mathcal{Q}; k), k}.$$

Here

$$C_0(\mathcal{P}, \mathcal{Q}; k) = \mathcal{P} \circ \mathcal{Q}.$$

Notice that Theorem 1.5 recovers the Toeplitz quantization for functions [BMS94].

This new framework suggests several directions for future research. It may allow one to construct quantization on *CR* and *contact* manifolds or in *sub-Riemannian* settings, to investigate star-product structures beyond functions, and to explore connections with quantum field theory models where both position and momentum variables are essential. We are also investigating more effective ways to represent the resulting formal power series and to relate its coefficients to geometric invariants.

2 Current Work

My present work focuses on extending the results described above by relaxing technical assumptions and deepening the asymptotic expansions of Bergman and Toeplitz kernels.

In particular, while the previous expansions assume a compact Kähler manifold and a positive line bundle, I am now investigating the case of manifolds with finite volume or more general curvature conditions. This extension is motivated both by mathematical interest and by applications in mathematical physics, where non-compact or singular geometries naturally arise.

At the same time, I am computing the higher-order coefficients of the Bergman kernel. These coefficients encode important geometric information. For instance, the leading term controls lower bounds that are used in the proof of the Donaldson-Sun theorem [DS14], a theory I am currently studying in parallel. The second term involves the scalar curvature of the manifold, while in the Toeplitz operator setting, the corresponding term reflects the Laplacian or Poisson structure associated with the line bundle. Understanding these terms is also closely connected to Donaldson's conjectures on projective manifolds and to broader questions in complex geometry concerning the geometric meaning of higher Bergman invariants.

Technically, this work involves several intertwined challenges. First, adapting the stationary phase and scaling methods to non-compact or finite-volume manifolds requires careful control of boundary behavior and asymptotic decay. Second, computing the higher coefficients of the Bergman kernel demands precise manipulation of adjoint operators, local expansions, and curvature terms. Finally, in the semiclassical Toeplitz pseudodifferential setting, although the Poisson bracket structure of the operators is already apparent at the next order, a systematic formulation of a full semiclassical quantization theorem remains in progress. This involves finding a natural framework for the pseudodifferential operator analogue of the Berezin-Toeplitz quantization, which can describe the formal power series and higher-order coefficients in a clear and general way.

These ongoing projects not only deepen and generalize my previous results but also lay the technical foundation for my future research. In particular, the tools and methods I am developing, explicit coefficient computations, semiclassical expansions, and Poisson bracket structures, will be essential for studying Bergman and Szegő kernels, Berezin-Toeplitz quantization, and pseudodifferential operators in *CR*, *contact*, and *sub-Riemannian* geometries. Moreover, this line of work provides a bridge between complex geometry and microlocal analysis, preparing for investigations into Fourier integral operators, deformation quantization, and geometric quantization in broader contexts.

3 Future Research Directions(PhD Proposal)

In addition to extending my previous and current research, I am eager to explore several new directions. The first one is the Cauchy-Riemann(CR) setting of projection and kernel theory. Also, we are curious about the contact and conformal Geometry. Last but not least, how to bridge the microlocal analysis tool into different geometric settings is also very stimulating and developmental.

3.1 Cauchy-Riemann Geometry and Szegő Kernel

The Szegő kernels are not merely technical tools but central objects in several branches of analysis and geometry. As reproducing kernels for CR functions, they link complex geometry, microlocal analysis, and quantization, and their asymptotics encode deep information about curvature, spectral invariants, and even boundary phenomena in mathematical physics. In particular, the Szegő kernel can be seen as the boundary analogue of the Bergman kernel, arising naturally on strictly pseudoconvex CR manifolds and on the boundaries of complex domains. Despite its long history, key aspects of its full asymptotic expansion remain only partially understood, especially in non-integrable or higher codimension settings, leaving substantial room for new analysis and new invariants.

Building on my previous investigations into the asymptotics of the Bergman kernel and its applications to geometric quantization, I intend to develop a comprehensive analytic study of the Szegő kernel on CR manifolds. A key operator underlying the Szegő kernel is the Kohn Laplacian. Classic works of Hörmander [Hör67] and Boutet de Monvel [Bou74] established hypoellipticity and subelliptic estimates, but explicit parametrices and semiclassical structures remain largely undeveloped. Constructing explicit parametrices and symbolic expansions for the Kohn Laplacian in model situations will be an underlying work throughout the whole research.

More precisely, let M be a CR manifold and $\square_b^{(q)}$ the Kohn Laplacian acting on $(0, q)$ -forms. Under the $Y(q)$ -condition one can construct a parametrix $N^{(q)}$ of $\square_b^{(q)}$, whereas if the $Y(q)$ -condition fails the analysis shifts naturally to the Szegő kernel

$$S^{(q)} : L^2(M) \rightarrow \ker \square_b^{(q)},$$

whose Fourier integral operator structure is described in [BS75; Hsi08]. As $\mathbb{C}^n$ provides the model case for the Bergman kernel, the Heisenberg group plays an analogous role for the Szegő kernel. Thus, explicit formulas for $N^{(q)}$ and $S^{(q)}$ in this model are therefore crucial for understanding the general situation. Furthermore, for a CR line bundle $L \rightarrow M$ one can define the Kohn Laplacian $\square_{b,k}^{(q)}$ on L^k together with its parametrix $N_k^{(q)}$ and Szegő projection $S_k^{(q)}$. In the high tensor powers $k \rightarrow \infty$, one expects a CR scaling behavior such that $N_k^{(q)}$ and $S_k^{(q)}$ converge, after appropriate rescaling, to their model analogues on the Heisenberg group.

My immediate objective is to employ semiclassical techniques to derive explicit formulas or asymptotics for the Szegő kernel and associated Toeplitz operators on the Heisenberg group equipped with quadratic potential and weight functions. At this first stage, the emphasis will be on capturing the leading order behavior of these objects, which should already illuminate how the geometry and the potential interact. Building on this foundation, a second goal is to extend these leading term results to more general CR manifolds and to develop refinements of the stationary phase method that are sensitive to the CR structure, thereby enabling the computation of subsequent terms and, ultimately, full expansions. In the longer term, I plan to integrate this analysis with the microlocal theory of pseudodifferential operators, even Fourier integral operators on CR manifolds, with the broader aim of exploring quantization phenomena arising from the Szegő kernel and Toeplitz operators in this geometric setting. This program not only advances the asymptotic analysis of kernels but also seeks to reveal deeper connections between CR geometry, semiclassical analysis, and geometric quantization.

In parallel with the semiclassical analysis, the Szegő kernel admits a Fourier integral

operator expansion of the form

$$S(x, y) = \int e^{it\varphi(x, y)} (t^n a_0(x, y) + t^{n-1} a_1(x, y) + \cdots).$$

For Toeplitz operators, i.e. operators of the form $T_{\mathcal{P}} = S\mathcal{P}S$ with $\mathcal{P}$ a pseudodifferential or Fourier integral operator, one expects a similar asymptotic expansion

$$T_{\mathcal{P}}(x, y) = \int e^{it\varphi(x, y)} (t^{n+m} b_{\mathcal{P},0}(x, y) + t^{n+m-1} b_{\mathcal{P},1}(x, y) + \cdots).$$

The study of the coefficients of the Szegő projection is a classical and important subject in several complex variables and CR geometry. Determining the coefficients a_j or $b_{\mathcal{P},j}$ explicitly, describing their recurrence relations, and interpreting their CR or conformal geometric significance are natural and challenging problems.

3.2 Contact and Conformal Geometry

Contact and conformal geometry arise naturally in the study of complex manifolds, CR manifolds, and geometric quantization. Both play important roles in geometric analysis and mathematical physics. Contact structures generalize the boundary geometry of complex domains to a much broader class of odd-dimensional manifolds, while conformal structures provide a natural setting for studying angle-preserving structures and invariant operators. Together, they provide a rich arena in which ideas developed for kernel functions and quantization in complex and CR settings can be tested, modified, and generalized.

I aim to develop a unified analytic framework for defining Szegő-type and Toeplitz-type operators in both contact and conformal geometric settings. In contact geometry, such operators are traditionally constructed on manifolds equipped with CR structures and Hardy space projections, where they encode microlocal and quantization data. I seek to extend these constructions to more general contact manifolds, exploring their symbolic behavior, spectral properties, and connections to contact invariants. In particular, I am interested in the study of sub-Riemannian Laplacians as natural generalizations of the Kohn Laplacian. By adapting techniques from CR geometry, such as symbolic calculus, microlocal analysis, and heat kernel methods, to the contact setting, I aim to construct parametrices, obtain asymptotic expansions for heat kernels, and investigate spectral properties and trace formulas that reflect underlying contact invariants. This approach may also allow the definition of generalized Szegő-type and Toeplitz-type operators on contact manifolds without integrable CR structures, revealing new analytic tools for probing geometric rigidity and invariants in odd-dimensional manifolds.

More precisely, we aim to identify suitable conditions under which Boutet de Monvel's construction of the parametrix of the Kohn Laplacian can be adapted to the sub-Riemannian Laplacian. In parallel, developing a contact-geometric analogue of Boutet-Sjöstrand's Fourier integral operator description of the Szegő kernel is another natural and compelling problem.

In conformal geometry, I plan to develop analogues of the Szegő limit theorem and associated Toeplitz operators in settings governed by angle-preserving structures. My focus is on constructing conformally invariant kernels and investigating their semiclassical expansions, symbolic calculus, and spectral asymptotics. By generalizing classical results on Szegő kernels and Toeplitz operators to conformal manifolds, I hope to uncover new

invariants arising from curvature and conformal structure. This line of research may also connect with developments in geometric analysis, including conformal anomalies, functional determinants, and the study of critical metrics, providing a bridge between kernel analysis, quantization, and conformal invariants.

A particularly concrete direction is to analyze the coefficients $a_j(x, y)$ appearing in the Fourier integral expansion of the Szegő kernel and their analogues for Toeplitz operators. Understanding how these coefficients transform under conformal changes and extracting their invariant content could lead to new conformal invariants and clarify the geometric meaning of the semiclassical terms.

Through these studies, I aim to deepen the interplay between analysis, geometry, and quantization, and to uncover new geometric and spectral invariants in both contact and conformal settings.

3.3 Bridging Geometry and Microlocal Analysis

A central focus of my research is the study of operators that encode the essential microlocal and spectral information across complex, CR, contact, and conformal geometric settings. In particular, central operators, such as Kohn or sub-Riemannian Laplacians, play a pivotal role in capturing the interplay between geometry and analysis. Inspired by the symbolic frameworks of Hörmander or Boutet de Monvel, I aim to develop a tailored symbolic calculus that systematically tracks asymptotic expansions, commutator structures, and functional properties of these operators. This approach allows the construction of parametrices, heat kernels, and semiclassical approximations in a unified framework, extending techniques traditionally confined to CR or complex manifolds to more general contact and conformal settings.

A particularly promising direction is the development of a CR analogue of the Donaldson-Sun theory, whose impact in complex geometry has been profound but remains largely unexplored from a microlocal analytic viewpoint. Incorporating symbolic calculus and related operators into this setting might extend Donaldson-Sun-type results to a broader geometric context.

By interpreting geometric structures through microlocal and semiclassical lenses, I seek to reveal subtle spectral and symbolic phenomena that are invisible to purely geometric methods. This perspective facilitates the study of kernel expansions, pseudodifferential and Toeplitz operators, and their connections with the underlying Poisson or symplectic structures. Moreover, developing a unified semiclassical framework across these diverse geometric contexts opens the door to applying Fourier integral operator techniques, exploring deformation and geometric quantization, and establishing bridges between analytic constructions and geometric invariants.

Ultimately, this line of research aims to unify technical analytic tools with geometric intuition, uncovering deeper structural relationships among complex, CR, and contact geometries. By connecting kernel asymptotics, operator theory, and microlocal analysis, I hope to provide new insights into global geometric and dynamical problems, enrich the understanding of invariants in diverse geometric settings, and create analytic frameworks that are robust enough to address questions in both geometric analysis and mathematical physics.

4 Conclusion

I have a solid foundation in complex geometry and microlocal analysis, developed through my master's thesis on the Bergman kernel and Toeplitz quantization. This work has given me both the technical depth and the research experience needed to tackle advanced problems at the interface of analysis and geometry. In particular, I have begun to develop innovative methods for quantization and full expansions of the Bergman kernel, which I am eager to extend to related settings.

Among the faculty, Professor Nikhil Savale's research stands out as a perfect match. His work on the Bergman and Szegő kernels, as well as on contact and sub-Riemannian geometry and quantization, directly aligns with both my past projects and my planned future directions. Joining his group would allow me to acquire expertise in contact and conformal geometry and to refine my analytical skills, while also contributing fresh perspectives from my own approaches to kernel expansions and quantization. Moreover, Professors Dmitri Zaitsev and Andreea Nicoara are leading experts in CR geometry, with their innovative research, especially those related to the Kohn Laplacian and its estimates, offering an excellent environment to further develop this project.

This institution offers an environment where cutting-edge analysis and geometry interact, and it provides the intellectual community and geographic setting that will support my growth. Earning a PhD here is a crucial step toward my long-term goal of becoming an independent researcher with a strong international network, and I am confident that my background and ambitions make me a strong fit for this program and its collaborative research culture.

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